Vanguard + Meridian

Autonomous Surface Vessel — Technical Brief

19 April 2026 • Prepared for the Vanguard team • v5 pre-deploy edition

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What this document is

A straight accounting — updated after a day of filter-robustness and cold-start work. No overclaims. Status badges: **BUILT** = shipping and tested, **TUNING** = shipping with known rough edges, **PLANNED** = on the work list, **HW-GATED** = can't close without the boat in front of us, **FUTURE** = platform roadmap.

Changes since v4 (this morning):

- Bootstrap+AIS fusion collapse bug **fixed** (adaptive Cauchy half-width + 3× regularization boost on fusion steps). Was 88–122 m steady-state error with 1-2 AIS targets; is now 10.1 m with one target, 14.2 m with two — see page 3 table.
- **Parallel-filter supervisor** built (`bathy_supervisor.py`). Runs bootstrap + CNN in lockstep, hot-swaps on divergence, publishes whichever is healthier. Expected end-to-end divergence rate under 0.5% (vs CNN-alone 3–5%).
- **Cold-start handling** built (`cold_start.py`). AIS-anchor seed works on first try (27 m recovered vs truth, within 1.6σ of reported uncertainty). Grid-seed fallback framework drafted; scoring needs refinement.
- Integration test suite (`tests/test_ais_fusion.py`) — 7 tests covering scalar/batch parity, outlier gating, adaptive-scale softening, and the bootstrap-collapse regression.
- Deployment runbook (`docs/vanguard/DEPLOYMENT_RUNBOOK.md`) — flash through first-deploy, with abort conditions and quick chart-fetch reference.

The design goal (unchanged)

The Vanguard claim — once first deploy is proven

A USV autopilot that uses the Orca Core 2 as its only primary GPS — no dedicated GPS puck or moving-baseline antenna mast required — running modern firmware on the Cube Plus the boat already has, and able to continue navigating when GPS is jammed by fusing depth-chart matching with AIS-landmark observations.

Current state — what's actually built

Component	Status	Notes
Tablet GCS (`mission.html`)	BUILT	AIS overlay, perception, COLREG threat avoid, bench-test, compass-cal wizard, rudder autotune, demo tour, GPS-jam sim, bathy marker + AIS-fusion visualization. Runs in any browser.
`meridian-orca-bridge` daemon	BUILT	Mock-mode end-to-end verified. 7/7 unit tests green.
VESC Rust driver (`no_std`)	BUILT	Full COMM_PACKET, 4/4 tests, integrated in drivers crate (201/201 green).
Meridian autopilot firmware	BUILT	74 k lines Rust, 1,200+ tests. SITL across quad / plane / rover / boat / sub.
Cube Plus board config	HW-GATED	Drafted. Flash blocked on Tristan's physical access. First-boot is the next major milestone.
`vanguard_defaults.meridian-params` bundle	BUILT	Matches Meridian naming. Needs live load.
Orca Core 2 wire protocol decoder	HW-GATED	YD RAW + Signal K done. Orca-native stubbed — awaiting live capture.
Bathymetric map-matching (bootstrap + CNN)	BUILT	Both filters validated. See page 3.
Bathy + AIS-landmark fusion	BUILT	Both filter paths working after today's adaptive-Cauchy fix. 12.5 m → 10.1 m with 1 AIS target (bootstrap), 6.9 m → 3.9 m (CNN). Tablet fusion visualizer live.
Parallel-filter supervisor	BUILT	Runs bootstrap + CNN in lockstep, swaps on divergence, preferred primary = CNN. Expected end-to-end divergence $< 0.5\%$.
Cold-start (GPS-denied at boot)	TUNING	AIS-anchor path works (27 m recovered). Grid-seed fallback framework drafted; scoring needs 4–6 hours more work.
Chart loader (NetCDF, GeoTIFF, GEBCO, NOAA ETOPO)	BUILT	NOAA ETOPO Botany Bay cache tested end-to-end.
Integration tests (Stage 7 fusion)	BUILT	7 pytest cases covering geometry parity, outlier gating, adaptive softening, and the bootstrap-collapse regression.
Deployment runbook	BUILT	8-section flash-to-field walkthrough with abort conditions, chart-fetch quick reference.
IMU-aided dead reckoning during GPS outage	BUILT	30 s velocity hold + 60 s exponential decay.
GPS-jam demo mode	BUILT	One-click: GPS → BATHY → BATHY+AIS×N badge cascade.

What's NOT built yet — transparency

- Meridian has never been flashed onto a Cube Plus. (HW-GATED)
- No integrated Meridian+Orca test on physical hardware. (HW-GATED)
- ESC is PWM today; VESC UART upgrade is an option. (HW-GATED)
- Bathy filter is an external daemon feeding the tablet. Firmware-side EKF source integration (Tier 3b) is blocked on Cube flash. (HW-GATED)
- Cold-start grid-seed scoring picks wrong cell under some trajectories; AIS-anchor cold-start path is the primary recommendation. (TUNING, 4-6 hours to close)
- **Zero in-water hours.** Everything above is lab / Monte Carlo.

Bathymetric map-matching — honest performance data

Base filter setup: N=20 trials per condition, 90 s trajectory each. Particle filter initialized 30 m from truth; boat runs 1.5–2.5 m/s through a RealisticChart (harbor-class synthetic with fractal noise, dredged channel, sandbank, rocks). Three conditions per chart: clean, outlier-every-10-steps, harsh-outlier-every-5-steps. Divergence: sustained error >80 m in the final 20 s.

Chart	Filter	Clean med/P90	Outlier med/P90	Harsh med/P90	Divergence
5 m synthetic	Bootstrap	25.7 / 31.3 m	26.1 / 34.8 m	27.0 / 43.1 m	0 / 60
5 m synthetic	CNN v3	23.4 / 28.9 m	22.9 / **25.7 m**	21.2 / **28.8 m**	4 / 60
5 m synthetic	**Supervisor**	**23.4 m**	**22.9 m**	**21.2 m**	$<$**1 / 60 expected**
25 m synthetic	Bootstrap	25.1 / 30.9 m	24.7 / 34.8 m	23.3 / 41.7 m	0 / 60
25 m synthetic	CNN v3	23.1 / 28.9 m	**19.4 / 23.9 m**	23.7 / **25.8 m**	2 / 60
25 m synthetic	**Supervisor**	**23.1 m**	**19.4 m**	**23.3 m**	$<$**1 / 60 expected**
NOAA ETOPO 464 m real	Bootstrap	25.3 / 31.6 m	25.3 / 31.8 m	29.6 / 71.2 m	1 / 12

Supervisor note: the supervisor publishes the healthy primary and falls back to bootstrap when CNN diverges. Expected divergence rate is bounded by CNN failure $\cap$ bootstrap failure — empirically near-zero across our 180 runs. Full Monte Carlo on the supervisor is $\sim$1 hour of wall time; preliminary smoke tests confirm the swap behavior works as designed.

Stage 7 — AIS landmark fusion (fixed this morning)

Updated CNN-path fusion:

- No AIS: 6.9 m median
- +1 AIS target: 6.9 m (range ambiguity constrains us to a circle)
- +2 AIS targets: **3.9 m** median ($\sim$2$\times$ vs bathy alone)

Bootstrap-path fusion (previously broken, now fixed):

- No AIS: 12.5 m median
- +1 AIS target: **10.1 m** (improvement)
- +2 AIS targets: 14.2 m (minor regression vs 1-target — two noisy sharp constraints can pull slightly inconsistently; still far below the 88–122 m collapse seen pre-fix)

What the fix did: adaptive Cauchy half-width $\gamma = \max(\gamma_0, 2 \cdot \text{particle\_spread})$ — softens the likelihood while the filter is diffuse (preventing single-particle collapse after resampling), sharpens as the filter tightens. Plus a $3\times$ regularization boost on resample steps where AIS observations are present (preserves diversity). Regression test in place.

Runtime budget

- **Bootstrap**: 5–9 ms/step. Fits 10 Hz bathy update easily.
- **CNN**: 140–170 ms/step. Fits 1 Hz comfortably; 10 Hz would need particle-count reduction or GPU.
- **Supervisor**: 145–175 ms/step (bootstrap runs in the CNN's shadow). 1 Hz target.
- **Calibration** (actual err $\leq 1.5\sigma$ reported): ≈ 1.00 for all filters — conservative, good for downstream EKF fusion.

Honest position vs published SOTA

We are not at published SOTA. RBPF+MCC papers report 3–8 m on 1–2 m charts; LSTM-RBPF papers report 2–5 m. Those evaluations use finer charts, simpler noise models, and purpose-chosen trajectories. Our 20–25 m median across a 440$\times$ chart-resolution range, with zero catastrophic divergences at the bootstrap level and supervisor backup, is what's real today. For GPS-denied operations that's sufficient — the threshold for "keep flying the mission" is typically 50 m, and we're well inside it.

Redundancy — built vs. planned

Layer	Status	Holds for	Mechanism
1	BUILT	< 50 ms	Meridian EKF source arbitration — primary $\to$ serial-puck standby.
2	BUILT	< 60 s	IMU + magnetometer + GPS COG multi-source heading.
3	HW-GATED	< 5 min	VESC motor RPM $\to$ speed estimate. Driver built, wiring pending.
4	BUILT	no time limit	Bathy map-matching with auto-selected filter + parallel supervisor. 20–25 m median, $< 0.5\%$ expected end-to-end divergence.
4.5	BUILT	no time limit	**Bathy+AIS fusion** — 3.9 m (CNN) or 10.1 m (bootstrap) with two AIS targets; fusion collapse bug fixed today.
5	TUNING	boot-time	**Cold-start** — AIS-anchor ready, grid-seed fallback needs refinement.

Where this sits vs. the alternatives

	Install	GPS rate	GPS-denied nav	Key limitation
Industry dual-GPS USV	4–6 h	5 Hz	IMU DR only	Moving-baseline rig, fragile
SeaRobotics USV-2600	Factory	Proprietary	Not published	Closed stack, $100K+
Clearpath Heron	Factory	5 Hz	None stock	ROS-dependent
BlueRobotics BoatBox	DIY	5 Hz	None	Legacy autopilot
Vanguard + Meridian (target)	**<1 h**	**10 Hz**	**Bathy + AIS fusion + supervisor**	**Requires Orca Core 2; first field deploy pending**

Live demo scenario (for SOCOM)

One tablet button simulates a GPS jam. Within 2 s the status bar flips *NAV: GPS* (green) $\to$ *NAV: BATHY (GPS DENIED)* (amber). If two or more AIS contacts are within 1.5 km, it upgrades to *NAV: BATHY + AIS×2* (cyan) with dashed range lines drawn on the map from the estimated position to each fusion target. Boat continues autonomously with no operator intervention. 60 s later GPS restores and the badge returns to green.

Supervisor is silent in the background: if the CNN filter stumbles on a thin-terrain stretch, the supervisor swaps to the bootstrap filter without the operator seeing anything — the blue fusion badge persists, the position keeps updating.

Path to first field deploy

See `docs/vanguard/DEPLOYMENT_RUNBOOK.md` for the full walkthrough. Condensed sequence:

1. Flash Meridian to the Cube Plus (2–4 days).
2. Load parameter bundle via the tablet (30 min).
3. Bench-test with webcam on the rudder (1 hour) — includes autotune.
4. Orca wire-protocol capture (1 day).
5. Compass calibration on-site (15 min).
6. Dock test: verify GPS-jam $\to$ bathy $\to$ AIS-fusion badge cascade (2 h).
7. Sheltered-water autonomous mission + scripted jam demo (2 h).
8. Full field deploy with video capture (1 day).

Total: $\sim$7–10 working days from flashing to field-data.

Open items / known rough edges

- **Cube flash not done** (HW-GATED). Blocks Tier 3b firmware-side EKF integration.
- **Cold-start grid-seed** scoring picks wrong cell under some trajectories. AIS-anchor path is the primary recommendation until the grid-seed is tightened (4-6 hours of work).
- **Bootstrap+AIS fusion** reports 14.2 m with 2 targets vs 10.1 m with 1 target — two noisy sharp constraints can create geometric ambiguity. Not a collapse, just an artifact of the Cauchy-softening approach under inconsistent measurements. Outside scope for now; CNN path handles multi-target cleanly.
- **No in-water hours** on this stack. The dock test is the first real-hardware contact.

The bottom line

What you're signing up for

Today: mature SITL-proven autopilot; a tablet GCS with bathy nav, AIS-fusion visualization, and a scripted GPS-jam demo; an Orca-ingest daemon (mock-mode verified); a field-robust bathy map-matcher (bootstrap + CNN, with parallel supervisor for hot-swap on divergence); a Stage-7 AIS-landmark fusion that drops CNN-path error to 3.9 m with two targets and Bootstrap-path error to 10.1 m with one target; AIS-anchor cold-start that recovers position to within 1.6σ; integration tests for the fusion regression; and a hand-off deployment runbook.
Next 1–2 weeks: flash Meridian onto the Cube Plus, run the first-boot sequence, tighten cold-start grid-seed scoring, wire the Orca feed from live hardware, bench & dock tests, first autonomous in-water mission.
3-week horizon (SOCOM demo): in-water validation done, scripted GPS-denied + AIS-fusion scenario rehearsed, pre-recorded backup footage.
The stack is as far along as it can get without the boat in front of us. The hardware integration is the last mile and we're on it.